1. A rigid structural framework composed of slender members joined at their ends to span large distances is called a:

(A) Truss
(B) Portal Frame
(C) Continuous Beam
(D) Retaining Wall

2. The fundamental geometric building unit that gives planar trusses their inherent geometric rigidity is the:

(A) Triangle
(B) Rectangle
(C) Hexagon
(D) Trapezoid

3. In planar pin-jointed trusses, the relationship between number of members m and number of joints j for a perfect truss is:

(A) m = 2 * j - 3
(B) m = 2 * j + 3
(C) m = 3 * j - 2
(D) m = j + 3

4. A truss having fewer members than required for static determinacy (m < 2*j - 3) is classified as:

(A) Deficient (Unstable / Mechanism)
(B) Redundant
(C) Perfect
(D) Super-stable

5. A truss possessing more members than required by static determinacy (m > 2*j - 3) is termed:

(A) Redundant (Statically Indeterminate)
(B) Deficient
(C) Hypostatic
(D) Unstable

6. In ideal structural truss analysis, the connection joints between members are assumed to be:

(A) Frictionless spherical pins / hinges
(B) Fully rigid welded moment connections
(C) Semi-rigid elastomeric pads
(D) Sliding telescopic joints

7. In ideal truss theory, all external active loads and support reactions MUST act strictly at:

(A) The pin joints (nodes)
(B) The mid-span of individual members
(C) The quarter-points of top chord
(D) Random locations

8. Because all joints are frictionless pins and loads act solely at joints, every ideal truss member behaves as a:

(A) Two-Force Member (pure axial tension or compression)
(B) Beam-Column with bending and torsion
(C) Three-force shear member
(D) Flexural plate

9. A truss member subjected to equal and opposite pulling forces directed away from its pin ends is in:

(A) Axial Tension
(B) Axial Compression
(C) Transverse Bending
(D) Pure Torsion

10. A truss member subjected to pushing forces directed toward its pin joints is in:

(A) Axial Compression
(B) Axial Tension
(C) Pure Shear
(D) Hydrostatic Tension

11. The Method of Joints is based on analyzing the static equilibrium of each joint treated as a:

(A) Concurrent coplanar force system (Sigma Fx = 0, Sigma Fy = 0)
(B) Non-concurrent force system with moments
(C) Three-dimensional spatial continuum
(D) Dynamic vibration node

12. When applying the Method of Joints to a 2D planar truss, the maximum number of unknown member forces solvable at a single joint is:

(A) Two (due to 2 independent equations Sigma Fx = 0, Sigma Fy = 0)
(B) Three
(C) Four
(D) One only

13. When two non-collinear members meet at an unloaded joint with no external forces, the internal forces in both members are:

(A) Zero in both members
(B) Equal and compressive
(C) Equal and tensile
(D) Infinite

14. If three members meet at an unloaded joint where two members are collinear, the force in the third non-collinear member is:

(A) Zero
(B) Equal to the sum of the other two
(C) Compressive
(D) Tensile

15. If two non-collinear members meet at a joint where an external load is collinear with one member, the other member force is:

(A) Zero
(B) Equal to the external load
(C) Compressive
(D) Half the load

16. The Method of Sections is based on cutting an imaginary section through the truss and applying:

(A) Equilibrium equations of a non-concurrent rigid body (Sigma Fx = 0, Fy = 0, M = 0)
(B) Hooke's law of elasticity
(C) Energy conservation integrals
(D) Euler buckling criteria

17. When cutting a truss using the Method of Sections, the cut section should generally pass through no more than:

(A) Three unknown members
(B) Two unknown members
(C) Five unknown members
(D) One member only

18. In the Method of Sections, to find the force in a specific member directly, moments are conveniently taken about:

(A) The point of concurrency of the other two cut unknown members
(B) The center of mass of the truss
(C) The midspan of the cut member
(D) An external arbitrary support

19. A bridge truss having alternating diagonal members forming equilateral or isosceles triangles is known as a:

(A) Warren Truss
(B) Pratt Truss
(C) Howe Truss
(D) Fink Truss

20. A parallel chord truss where diagonal web members slope towards the center in tension under gravity loads is a:

(A) Pratt Truss
(B) Howe Truss
(C) K-Truss
(D) Bowstring Truss

21. A parallel chord truss where diagonal web members slope away from the center in compression under gravity loads is a:

(A) Howe Truss
(B) Pratt Truss
(C) Warren Truss
(D) Vierendeel Truss

22. A roof truss characterized by subdivided triangular bays suitable for steep residential and warehouse pitches is the:

(A) Fink Truss
(B) King Post Truss
(C) Queen Post Truss
(D) Baltimore Truss

23. A simple roof truss composed of a single vertical central tie post with two inclined rafters is a:

(A) King Post Truss
(B) Queen Post Truss
(C) Pratt Truss
(D) Howe Truss

24. A roof truss featuring two vertical posts framing a flat central tie beam is a:

(A) Queen Post Truss
(B) King Post Truss
(C) Warren Truss
(D) Lattice Truss

25. In a simply supported bridge truss carrying downward gravity deck loads, the top chord members are universally in:

(A) Axial Compression
(B) Axial Tension
(C) Zero Force
(D) Pure Bending

26. In a simply supported bridge truss carrying downward gravity deck loads, the bottom chord members are universally in:

(A) Axial Tension
(B) Axial Compression
(C) Zero Force
(D) Pure Torsion

27. In a cantilever truss extending horizontally from a fixed vertical wall carrying downward gravity loads at its tip, top chord members are in:

(A) Axial Tension
(B) Axial Compression
(C) Zero Force
(D) Pure Shear

28. In the cantilever truss of Q27, bottom chord members under downward gravity loads are in:

(A) Axial Compression
(B) Axial Tension
(C) Zero Force
(D) Torsion

29. A rigid frame possessing rigid moment connections instead of pin joints and carrying loads via bending moments is a:

(A) Vierendeel Frame / Rigid Frame
(B) Pin-jointed Truss
(C) Cable Net Structure
(D) Tensegrity Mast

30. The presence of zero-force members in practical engineering trusses is essential because they:

(A) Provide lateral bracing, buckling restraint, and handle alternate load cases (wind/quake)
(B) Increase dead weight needlessly
(C) Reduce structural efficiency
(D) Violate static determinacy

31. For a 2D truss with 5 joints (j = 5), the number of members required for a statically determinate perfect truss is:

(A) 7 (m = 2*5 - 3 = 7)
(B) 10
(C) 8
(D) 6

32. For a 2D truss with 8 joints (j = 8), the number of members for a perfect truss is:

(A) 13 (m = 2*8 - 3 = 13)
(B) 16
(C) 11
(D) 14

33. If a truss has j = 6 joints and m = 10 members, its degree of static indeterminacy (internal redundancy) is:

(A) 1 (Redundant to degree 1, since 10 - (2*6 - 3) = 1)
(B) 0 (Determinate)
(C) 2
(D) 3

34. If a truss has j = 7 joints and m = 10 members, it is:

(A) Deficient (Unstable, since 10 < 2*7 - 3 = 11)
(B) Perfect
(C) Indeterminate
(D) Statically locked

35. In graphical truss analysis, the closed vector polygon diagram combining all joint force vectors is known as:

(A) Maxwell-Cremona Diagram
(B) Mohr's Circle Diagram
(C) Williot-Mohr Diagram
(D) Influence Line Diagram

36. Bow's Notation in graphical structural analysis designates forces and members by:

(A) Letters assigned to the spaces between forces in the space diagram
(B) Sequential Roman numerals
(C) Cartesian vector subscripts
(D) Euler angles

37. A space truss is a three-dimensional structural framework where the internal determinacy criterion is:

(A) m = 3 * j - 6
(B) m = 2 * j - 3
(C) m = 3 * j - 3
(D) m = 6 * j - 12

38. The tetrahedron is the basic geometrically rigid three-dimensional unit of a:

(A) Space Truss
(B) Planar Truss
(C) Cable Stayed Bridge
(D) Suspension Catenary

39. At any joint in a space truss, the number of independent scalar equilibrium equations available is:

(A) Three (Sigma Fx = 0, Sigma Fy = 0, Sigma Fz = 0)
(B) Two
(C) Six
(D) One

40. A compound truss is formed by connecting two or more simple trusses together using:

(A) Three non-parallel non-concurrent links, or a common joint and a link
(B) Welded continuous beams
(C) Rigid foundation slabs
(D) Pure torque tubes

41. An equilateral triangular truss ABC (sides 4 m) has roller at B and hinge at A. A vertical load W = 60 kN acts at top apex C. Support reactions are:

(A) 30 kN at A, 30 kN at B
(B) 60 kN at A, 0 kN at B
(C) 45 kN at A, 15 kN at B
(D) 20 kN at A, 40 kN at B

42. In Q41 (apex load 60 kN, equilateral triangle), the force in member AC is:

(A) 34.64 kN Compression
(B) 34.64 kN Tension
(C) 30.00 kN Compression
(D) 60.00 kN Tension

43. In Q41 (apex load 60 kN, equilateral triangle), the force in bottom member AB is:

(A) 17.32 kN Tension
(B) 17.32 kN Compression
(C) 30.00 kN Tension
(D) 34.64 kN Tension

44. In Q41, the force in member BC is:

(A) 34.64 kN Compression
(B) 34.64 kN Tension
(C) 30.00 kN Compression
(D) 17.32 kN Compression

45. A simple symmetric roof truss has span 12 m and height 3 m. The slope angle theta of the rafters is:

(A) 26.57 degrees (tan(theta) = 3 / 6 = 0.5)
(B) 30.00 degrees
(C) 45.00 degrees
(D) 14.04 degrees

46. A right-angled triangular bracket truss ABC has horizontal member AB = 3 m, vertical wall member BC = 4 m, and diagonal AC = 5 m. A vertical load of 12 kN acts at joint A. Force in AB is:

(A) 9 kN Tension
(B) 9 kN Compression
(C) 15 kN Tension
(D) 12 kN Compression

47. In Q46 (bracket truss with 12 kN at A), the force in diagonal member AC is:

(A) 15 kN Compression
(B) 15 kN Tension
(C) 9 kN Compression
(D) 20 kN Compression

48. In Q46, the force in vertical wall member BC is:

(A) 12 kN Tension
(B) 12 kN Compression
(C) 15 kN Compression
(D) 0 kN

49. A planar truss has joint D with members DA, DB, DC. DA and DC are collinear along the horizontal. If no external load acts at D, the force in member DB is:

(A) 0 kN (Zero-force member)
(B) Equal to force in DA
(C) Tensile
(D) Compressive

50. In Q49, the forces in the two collinear members DA and DC must be:

(A) Equal in magnitude and of the same sense (both T or both C)
(B) Equal in magnitude and opposite sense
(C) Zero
(D) Unrelated

51. At an unloaded joint E, only two members EA and EB meet at an angle of 70 degrees. The force in member EA is:

(A) 0 kN
(B) 10 kN Tension
(C) 5 kN Compression
(D) Indeterminate

52. In Q51, the force in member EB is:

(A) 0 kN
(B) 10 kN Compression
(C) 5 kN Tension
(D) Indeterminate

53. A Warren truss has span 18 m composed of 3 bays of 6 m. Height is 4 m. A vertical load of 100 kN acts at midspan bottom chord. The reaction at each support is:

(A) 50 kN
(B) 100 kN
(C) 25 kN
(D) 75 kN

54. In Q53, the diagonal web members have length L_d = sqrt(3^2 + 4^2) = 5 m. The sine of the diagonal angle is:

(A) 0.80 (sin(theta) = 4 / 5)
(B) 0.60
(C) 0.75
(D) 1.00

55. In Q53-54, the axial force in the end diagonal member connected to the support reaction of 50 kN is:

(A) 62.50 kN Compression
(B) 62.50 kN Tension
(C) 50.00 kN Compression
(D) 40.00 kN Compression

56. In Q53-55, the axial force in the bottom chord member at the support is:

(A) 37.50 kN Tension
(B) 37.50 kN Compression
(C) 50.00 kN Tension
(D) 62.50 kN Tension

57. In Q53-56, using the Method of Sections cut through the midspan, the force in the top chord member is:

(A) 75.00 kN Compression
(B) 75.00 kN Tension
(C) 50.00 kN Compression
(D) 100.00 kN Compression

58. A Pratt bridge truss of 6 bays (each 4 m wide, height 4 m) has total span 24 m. Total downward load of 120 kN acts at each bottom joint. End reaction is:

(A) 300 kN
(B) 360 kN
(C) 240 kN
(D) 600 kN

59. In a Pratt truss, the vertical web members under gravity deck loads are primarily in:

(A) Axial Compression
(B) Axial Tension
(C) Zero Force
(D) Bending

60. In a Pratt truss, the diagonal web members under gravity deck loads are primarily in:

(A) Axial Tension
(B) Axial Compression
(C) Zero Force
(D) Torsion

61. In a Howe truss, the vertical web members under gravity deck loads are primarily in:

(A) Axial Tension
(B) Axial Compression
(C) Zero Force
(D) Shear

62. In a Howe truss, the diagonal web members under gravity deck loads are primarily in:

(A) Axial Compression
(B) Axial Tension
(C) Zero Force
(D) Flexure

63. A cantilever truss extends 6 m from a wall with depth 3 m. A vertical load of 40 kN acts at its tip. The tension in the top horizontal tie member at the wall is:

(A) 80 kN Tension
(B) 40 kN Tension
(C) 120 kN Tension
(D) 60 kN Tension

64. In Q63, the compressive force in the bottom horizontal chord member at the wall is:

(A) 80 kN Compression
(B) 40 kN Compression
(C) 120 kN Compression
(D) 60 kN Compression

65. A planar truss has 9 joints and 15 members. The internal degree of indeterminacy is:

(A) 0 (Perfect truss, since 2*9 - 3 = 15)
(B) 1
(C) 2
(D) -1

66. A planar truss has 10 joints and 16 members. The truss is:

(A) Deficient by 1 member (Unstable mechanism, since 2*10 - 3 = 17)
(B) Perfect
(C) Indeterminate by 1
(D) Statically locked

67. In a symmetrically loaded Warren truss without verticals, alternate diagonals have forces that are:

(A) Equal in magnitude with alternating signs (one T, next C)
(B) All tensile
(C) All compressive
(D) Zero

68. At a joint with 4 members forming an X (two intersecting continuous straight members), the forces in opposite collinear members are:

(A) Equal in magnitude and of the same sense
(B) Zero
(C) Opposite in sign
(D) Independent

69. The primary advantage of the Method of Sections over the Method of Joints is that it allows:

(A) Direct calculation of an internal member force without solving preceding joints
(B) Analysis of statically indeterminate trusses
(C) Calculation of nonlinear deflection
(D) Inclusion of joint friction

70. A section cut in the Method of Sections divides the truss into two distinct bodies. Equilibrium requires:

(A) Either isolated part must satisfy Sigma Fx = 0, Sigma Fy = 0, Sigma M = 0
(B) Only the left side satisfies equilibrium
(C) The cut members must have zero force
(D) Sigma Fx + Sigma Fy = 0 only

71. A 45-degree diagonal brace in a 3 m x 3 m square truss panel carries a shear force of V = 30 kN. The axial force in the brace is:

(A) 42.43 kN
(B) 30.00 kN
(C) 21.21 kN
(D) 60.00 kN

72. In a pin-jointed truss, if an external concentrated load is accidentally placed at midspan of a member, the member will experience:

(A) Combined axial force and transverse bending moment (acting as a beam-column)
(B) Pure axial tension
(C) Pure shear without bending
(D) Zero stress

73. When analyzing space trusses using the Method of Joints, the number of unknown member forces that can be determined at any single joint is:

(A) Three (using Sigma Fx = 0, Sigma Fy = 0, Sigma Fz = 0)
(B) Two
(C) Six
(D) Four

74. Secondary stresses in real engineering trusses arise primarily due to:

(A) Rigidity (stiffness) of gusset plate connections and member self-weight
(B) Frictionless pins
(C) Pure axial loads
(D) Thermal expansion alone

75. Under standard code specifications (e.g., IS 800 / AISC), secondary bending stresses in trusses can be safely neglected if:

(A) Members are slender (L / r > 50) and lines of action intersect concentrically at joints
(B) Members are extremely thick
(C) The truss has zero joints
(D) Loads are purely dynamic

76. A 6-bay Pratt bridge truss has span L = 24 m (each panel 4 m) and height h = 4 m. Loads of 60 kN act at joints L1, L2, L3, L4, L5. Left reaction R_A is:

(A) 150 kN
(B) 180 kN
(C) 300 kN
(D) 120 kN

77. In Q76, the axial force in the end post diagonal member U1-L0 is:

(A) 212.13 kN Compression
(B) 150.00 kN Compression
(C) 212.13 kN Tension
(D) 180.00 kN Compression

78. In Q76, the axial force in bottom chord member L0-L1 is:

(A) 150.00 kN Tension
(B) 150.00 kN Compression
(C) 212.13 kN Tension
(D) 100.00 kN Tension

79. In Q76-78, the axial force in vertical member U1-L1 is:

(A) 0 kN (Zero-force member under bottom chord loading)
(B) 60 kN Tension
(C) 150 kN Compression
(D) 60 kN Compression

80. In Q76-79, using a cut section through panel 2 (cutting U1-U2, U2-L1, L1-L2), the force in top chord U1-U2 is:

(A) 150.00 kN Compression
(B) 200.00 kN Compression
(C) 250.00 kN Compression
(D) 120.00 kN Compression

81. In Q76-80, the vertical shear carried by panel 2 is V = R_A - 60 = 90 kN. The axial tension in diagonal U2-L1 is:

(A) 127.28 kN Tension
(B) 90.00 kN Tension
(C) 150.00 kN Tension
(D) 63.64 kN Tension

82. In Q76-81, the axial force in bottom chord member L1-L2 is:

(A) 150.00 kN Tension
(B) 240.00 kN Tension
(C) 210.00 kN Tension
(D) 180.00 kN Tension

83. In Q76-82, in panel 3 (midspan panel), the vertical shear is V = 150 - 60 - 60 = 30 kN. The tension in diagonal U3-L2 is:

(A) 42.43 kN Tension
(B) 30.00 kN Tension
(C) 60.00 kN Tension
(D) 21.21 kN Tension

84. In Q76-83, the maximum compression in the top chord occurs in member U2-U3 and equals:

(A) 210.00 kN Compression
(B) 150.00 kN Compression
(C) 240.00 kN Compression
(D) 180.00 kN Compression

85. In Q76-84, the maximum tension in the bottom chord occurs in member L2-L3 and equals:

(A) 210.00 kN Tension
(B) 150.00 kN Tension
(C) 240.00 kN Tension
(D) 180.00 kN Tension

86. A K-truss panel has height h = 6 m and width b = 4 m. A vertical shear of 120 kN acts across the panel. Each of the two diagonal members carries:

(A) 60 kN vertical shear (one in tension, one in compression)
(B) 120 kN shear
(C) 30 kN shear
(D) Zero shear

87. In Q86, the length of each K-truss diagonal is sqrt(2^2 + 3^2) = sqrt(13) = 3.606 m. The axial force in each diagonal is:

(A) 72.11 kN (one Tension, one Compression)
(B) 60.00 kN
(C) 120.00 kN
(D) 36.06 kN

88. A space truss tripod consists of three equal legs of length L = 5 m meeting at apex A at height h = 4 m. A vertical load of 120 kN acts at A. Force in each leg is:

(A) 50.00 kN Compression
(B) 40.00 kN Compression
(C) 60.00 kN Compression
(D) 30.00 kN Compression

89. In Q88, the horizontal spreading force transmitted to each of the three ground anchors is:

(A) 30.00 kN radially outward
(B) 40.00 kN
(C) 50.00 kN
(D) 15.00 kN

90. A Fink roof truss has 8 panels on a 16 m span with rise 4 m. Under symmetrical dead load of 10 kN per top joint, the force in the main horizontal tie is:

(A) 50.00 kN Tension
(B) 40.00 kN Tension
(C) 60.00 kN Tension
(D) 35.00 kN Tension

91. In an N-truss cantilever of 3 panels (each 2 m x 2 m), a tip load of 30 kN acts downward. The maximum tension in the top chord at the root wall is:

(A) 60 kN Tension
(B) 90 kN Tension
(C) 30 kN Tension
(D) 120 kN Tension

92. In Q91, the compressive force in the diagonal member at the wall panel is:

(A) 42.43 kN Compression
(B) 30.00 kN Compression
(C) 60.00 kN Compression
(D) 21.21 kN Compression

93. A symmetric planar truss has an internal counter-braced panel with two crossing diagonals. If diagonals are slender steel rods incapable of carrying compression:

(A) The compression diagonal buckles elastically and carries 0 force; tension diagonal carries all shear
(B) Both diagonals carry equal shear
(C) The truss collapses instantly
(D) The tension diagonal snaps

94. A bridge truss subjected to a moving load train experiences stress reversal in web members when:

(A) The resultant shear force changes sign from positive to negative as loads traverse
(B) Bending moment vanishes
(C) Total load doubles
(D) Temperature drops below zero

95. To accommodate stress reversal in web members without buckling, designers specify:

(A) Counters (tension-only cross-braces) or members designed for both tension and compression
(B) Zero-force rods
(C) Hinged cutouts
(D) Lead dampeners

96. A tower truss of square base 4 m x 4 m and height 20 m carries a horizontal wind shear of 40 kN at its top. The overturn overturning moment at base is:

(A) 800 kN*m
(B) 400 kN*m
(C) 160 kN*m
(D) 1600 kN*m

97. In Q96, the axial force induced in each of the two windward leg columns by overturning is:

(A) 100.0 kN Tension
(B) 200.0 kN Tension
(C) 50.0 kN Tension
(D) 400.0 kN Tension

98. In Q96-97, the axial force induced in each of the two leeward leg columns is:

(A) 100.0 kN Compression
(B) 200.0 kN Compression
(C) 50.0 kN Compression
(D) 400.0 kN Compression

99. A planar truss is internally indeterminate to degree 1. The redundant member force can be determined by applying:

(A) Castigliano's Second Theorem (Method of Minimum Strain Energy dU/dX = 0)
(B) Varignon's theorem
(C) Lami's theorem
(D) Archimedes principle

100. In computer structural analysis of large civil trusses, the fundamental matrix equation governing joint displacements is:

(A) [K] * {Delta} = {P} (where [K] is the global structural stiffness matrix)
(B) [M] * {P} = {Delta}
(C) [K] * {P} = {0}
(D) {Delta} = [K] * {P}^2

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Formulate the fundamental definition of an engineering truss, its four classical analytical assumptions, and the internal static determinacy criterion.

Q2. State and prove the three formal engineering rules for identifying Zero-Force Members in planar trusses by visual inspection.

Q3. Compare the Method of Joints and Method of Sections in detail, detailing their analytical procedures and relative advantages.

Q4. Analyze an equilateral triangular truss ABC of span 6 m (sides 6 m) supported by a hinge at A and roller at B, carrying a vertical load of 100 kN at apex C. Calculate all member forces using Method of Joints.

Q5. A Warren bridge truss has span 18 m (3 panels of 6 m, height 4 m). It carries a central downward load of 120 kN at midspan bottom joint L1. Calculate forces in top chord, bottom chord, and diagonals using Method of Sections.

Q6. A cantilever truss of span 6 m and depth 3 m carries vertical loads: 20 kN at mid-joint and 30 kN at free tip. Determine the wall reaction forces and member forces.

Q7. Explain the structural behavior and member force distributions in Pratt vs Howe bridge trusses under moving traffic loads.

Q8. A 6-bay Pratt truss of span 24 m (panel width 4 m, height 4 m) carries 80 kN loads at each bottom joint. Calculate the force in the center vertical member and diagonals.

Q9. A planar roof truss carries wind loads perpendicular to the rafter. Formulate the method for computing horizontal and vertical reactions and joint equilibrium.

Q10. Analyze a space tripod structure supported on three ball joints subjected to a vertical load W and horizontal load H at apex A.

Q11. Explain the Williot-Mohr diagram method for determining deflections of pin-jointed trusses graphically.

Q12. Derive the force in the tie rod of a King Post truss of span 10 m and rise 2.5 m under a central load of 50 kN.

Q13. Explain the concept of counter-bracing in bridge trusses subjected to moving train live loads.

Q14. Calculate the forces in a Fink truss rafter and tie under uniform vertical panel loads.

Q15. A planar truss has 6 joints and 9 members. Prove whether it is internally determinate, redundant, or deficient.

Q16. Derive the tension coefficient method for the analysis of 3D space trusses.

Q17. Explain the effect of temperature changes on statically determinate vs statically indeterminate trusses.

Q18. Calculate the horizontal displacement of the roller support of a simply supported symmetrical triangular truss under an apex load.

Q19. Describe the K-truss configuration and explain its aerodynamic and structural advantages in high-rise bridge towers.

Q20. Explain secondary stresses caused by gusset plate joint rigidity and state the design provisions for limiting them.

Q21. Calculate member forces in a 3-member cantilever bracket carrying a horizontal load at its apex.

Q22. Explain the Method of Substitute Members (Henrici's Method) for complex statically determinate trusses.

Q23. Analyze the effect of fabrication errors (initial lack of fit) in determinate and indeterminate trusses.

Q24. Discuss the design of transmission line electrical towers as 3D space trusses under wind and broken-wire conditions.

Q25. Explain why triangular panels prevent shear distortion while rectangular panels deform into parallelograms.

Q26. Derive the force in the vertical tie of a Queen Post truss of span 12 m with two panels of 4 m.

Q27. Explain the concept of an Influence Line Diagram (ILD) for a member in a bridge truss.

Q28. Calculate the axial forces in a tower truss subjected to lateral earthquake base shear.

Q29. Explain the Vierendeel truss and why it is not a true pin-jointed truss.

Q30. Summarize the complete structural design workflow for civil engineering roof and bridge trusses.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1: [A]
Q2: [A]
Q3: [A]
Q4: [A]
Q5: [A]
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Q97: [A]
Q98: [A]
Q99: [A]
Q100: [A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

SECTION B: DETAILED SOLUTIONS FOR SUBJECTIVE QUESTIONS

[illegible] ering truss, its four classical analytical assumptions, and the internal static determina

4. Civil Engineering Design & Stability Implications:
Compression members must be checked for Euler column buckling (P_cr = pi^2 * E * I / L_eff^2), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

>> Solution to Question 2:

[Problem Statement]: State and prove the three formal engineering rules for identifying Zero-Force Members in planar trusses by visual inspection.

- Internal determinacy satisfied via Maxwell's criterion: $m = 2*j - 3$.
- Joint node equilibrium confirmed: Vector resultant of all concurrent member forces vanishes at each pin.

4. Civil Engineering Design & Stability Implications:

Compression members must be checked for Euler column buckling ($P_{cr} = pi^2 * E * I / L_{eff}^2$), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

>> Solution to Question 3:

[Problem Statement]: Compare the Method of Joints and Method of Sections in detail, detailing their analytical procedures and relative advantages.

- Limitation: Limited to solving a maximum of 3 unknown forces per cut section.

3. Structural Equilibrium & Kinematic Compatibility Check:
- External determinacy verified via global equilibrium equations: Sigma Fx = 0, Sigma Fy = 0, Sigma M = 0.
- Internal determinacy satisfied via Maxwell's criterion: m = 2*j - 3.
- Joint node equilibrium confirmed: Vector resultant of all concurrent member forces vanishes at each pin.

4. Civil Engineering Design & Stability Implications:
Compression members must be checked for Euler column buckling (P_cr = pi^2 * E * I / L_eff^2), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

>> Solution to Question 4:

[Problem Statement]: Analyze an equilateral triangular truss ABC of span 6 m (sides 6 m) supported by a hinge at A and roller at B, carrying a vertical load of

3. Equilibrium of Joint B (Verification):
- Vertical equilibrium: Sigma Fy = 0:
R_B + F_BC * sin(60 deg) = 0 => 50 + F_BC * 0.8660 = 0 => F_BC = - 57.74 kN (Compression).
- Horizontal equilibrium: Sigma Fx = 0:
- F_AB - F_BC * cos(60 deg) = - 28.87 - (-57.74 * 0.5) = - 28.87 + 28.87 = 0 (Verified).

4. Summary of Member Forces:
- Member AC: 57.74 kN (Compression).
- Member BC: 57.74 kN (Compression).
- Member AB: 28.87 kN (Tension).

3. Structural Equilibrium & Kinematic Compatibility Check:
- External determinacy verified via global equilibrium equations: Sigma Fx = 0, Sigma Fy = 0, Sigma M = 0.
- Internal determinacy satisfied via Maxwell's criterion: m = 2*j - 3.
- Joint node equilibrium confirmed: Vector resultant of all concurrent member forces vanishes at each pin.

4. Civil Engineering Design & Stability Implications:
Compression members must be checked for Euler column buckling (P_cr = pi^2 * E * I / L_eff^2), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

>> Solution to Question 5:

Problem Statement: A Warren bridge truss has span 18 m (3 panels of 6 m, height 4 m). It carries a central downward load of 120 kN at midspan bottom joi

4 * F_{L0-L1} = 180 => F_{L0-L1} = + 45.0 kN (Tension).

3. Cut Section 2-2 through Panel 2 (cutting U1-U2, U2-L1, L1-L2):
Isolate left portion up to x = 8 m:
- Sum of Moments about Bottom Joint L1(x=6, y=0): Sigma M_{L1} = 0:
(R_L0 * 6.0 m) + (F_{U1-U2} * 4.0 m) = 0
(60 * 6.0) + 4 * F_{U1-U2} = 0
360 + 4 * F_{U1-U2} = 0 => F_{U1-U2} = - 90.0 kN (Compression of 90 kN in top chord).

- Sum of Vertical Forces: Sigma Fy = 0:
R_L0 - F_{U2-L1} * sin(theta) = 0
60 - F_{U2-L1} * 0.80 = 0 => F_{U2-L1} = + 75.0 kN (Tension of 75 kN in web diagonal).

- Sum of Moments about Top Joint U2(x=9, y=4): Sigma M_{U2} = 0:
- (R_L0 * 9.0 m) + (F_{L1-L2} * 4.0 m) = 0
- 540 + 4 * F_{L1-L2} = 0 => F_{L1-L2} = + 135.0 kN (Tension of 135 kN in bottom chord).

3. Structural Equilibrium & Kinematic Compatibility Check:
- External determinacy verified via global equilibrium equations: Sigma Fx = 0, Sigma Fy = 0, Sigma M = 0.
- Internal determinacy satisfied via Maxwell's criterion: m = 2*j - 3.
- Joint node equilibrium confirmed: Vector resultant of all concurrent member forces vanishes at each pin.

4. Civil Engineering Design & Stability Implications:
Compression members must be checked for Euler column buckling (P_cr = pi^2 * E * I / L_eff^2), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

>> Solution to Question 6:

[Problem Statement]: A cantilever truss of span 6 m and depth 3 m carries vertical loads: 20 kN at mid-joint and 30 kN at free tip. Determine the wall reaction

3 * R_Ax = - 300 kN*m => R_Ax = - 100.0 kN (100 kN Tension pulling away from wall).
- Horizontal Force Balance: Sigma Fx = 0:
R_Bx + R_Ax = 0 => R_Bx = + 100.0 kN (100 kN Compression pushing against wall).

3. Internal Member Forces at Root Panel:
- Top chord member AC carries the full horizontal tension: F_AC = 100.0 kN (Tension).
- Bottom chord member BD carries the full horizontal thrust: F_BD = 100.0 kN (Compression).
- Diagonal member BC carries the vertical shear of 50 kN:
Angle theta = atan(3 / 3) = 45 deg.
F_BC = 50 / sin(45 deg) = 50 * 1.4142 = 70.71 kN (Tension).

3. Structural Equilibrium & Kinematic Compatibility Check:
- External determinacy verified via global equilibrium equations: Sigma Fx = 0, Sigma Fy = 0, Sigma M = 0.
- Internal determinacy satisfied via Maxwell's criterion: m = 2*j - 3.
- Joint node equilibrium confirmed: Vector resultant of all concurrent member forces vanishes at each pin.

4. Civil Engineering Design & Stability Implications:
Compression members must be checked for Euler column buckling (P_cr = pi^2 * E * I / L_eff^2), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

>> Solution to Question 7:

[Problem Statement]: Explain the structural behavior and member force distributions in Pratt vs Howe bridge trusses under moving traffic loads.

Ideal for 19th-century timber-iron hybrid bridges where heavy wooden balks functioned well as compression diagonals, while vertical wrought iron rods served as tension ties.

3. Stress Reversal under Moving Wheel Loads:
As a train moves across the bridge, shear force in intermediate panels reverses sign. To prevent long tension diagonals from buckling in compression, designers install counter-braces (crossed diagonals).

3. Structural Equilibrium & Kinematic Compatibility Check:
- External determinacy verified via global equilibrium equations: Sigma Fx = 0, Sigma Fy = 0, Sigma M = 0.
- Internal determinacy satisfied via Maxwell's criterion: m = 2*j - 3.
- Joint node equilibrium confirmed: Vector resultant of all concurrent member forces vanishes at each pin.

4. Civil Engineering Design & Stability Implications:
Compression members must be checked for Euler column buckling (P_cr = pi^2 * E * I / L_eff^2), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

>> Solution to Question 6:

[Problem Statement: A 6-bay Pratt truss of span 24 m (panel width 4 m, height 4 m) carries 80 kN loads at each bottom joint. Calculate the force in the cente

- Horizontal members L2-L3 and L3-L4 are collinear and carry equal horizontal tension.
- By equilibrium of Joint L3 in vertical direction: Sigma Fy = 0:

F_{U3-L3} - 80 kN = 0 => F_{U3-L3} = 80.0 kN (Tension).

Conclusion: The center vertical in a deck-loaded Pratt truss acts as a vertical hanger in pure tension of 80 kN.

3. Structural Equilibrium & Kinematic Compatibility Check:

- External determinacy verified via global equilibrium equations: Sigma Fx = 0, Sigma Fy = 0, Sigma M = 0.
- Internal determinacy satisfied via Maxwell's criterion: m = 2*j - 3.
- Joint node equilibrium confirmed: Vector resultant of all concurrent member forces vanishes at each pin.

4. Civil Engineering Design & Stability Implications:

Compression members must be checked for Euler column buckling (P_cr = pi^2 * E * I / L_eff^2), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

>> Solution to Question 9:

[Problem Statement]: A planar roof truss carries wind loads perpendicular to the rafter. Formulate the method for computing horizontal and vertical reactions

Sigma M = 0.
- Internal determinacy satisfied via Maxwell's criterion: m = 2*j - 3.
- Joint node equilibrium confirmed: Vector resultant of all concurrent member forces vanishes at each pin.

4. Civil Engineering Design & Stability Implications:
Compression members must be checked for Euler column buckling (P_cr = pi^2 * E * I / L_eff^2), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

>> Solution to Question 10:

[Problem Statement]: Analyze a space tripod structure supported on three ball joints subjected to a vertical load W and horizontal load H at apex A.

>>[illegible] 50 kN.

at each pin.

4. Civil Engineering Design & Stability Implications:
Compression members must be checked for Euler column buckling (P_cr = pi^2 * E * I / L_eff^2), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

>> Solution to Question 14: [Problem Statement: Evaluate the forces in a Fink truss rafter and tie under uniform vertical panel loads. [illegible] subjected to moving live loads.

resist heavy wind and snow roof loads.

3. Structural Equilibrium & Kinematic Compatibility Check:
- External determinacy verified via global equilibrium equations: Sigma Fx = 0, Sigma Fy = 0, Sigma M = 0.
- Internal determinacy satisfied via Maxwell's criterion: m = 2*j - 3.
- Joint node equilibrium confirmed: Vector resultant of all concurrent member forces vanishes at each pin.

4. Civil Engineering Design & Stability Implications:
Compression members must be checked for Euler column buckling (P_cr = pi^2 * E * I / L_eff^2), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

>> Solution to Question 16: [illegible] determinate, redundant, or deficient.

members are ideal two-force axial elements carrying pure tension (+T) or compression (-C).

2. Mathematical Derivation & Step-by-Step Joint/Section Analysis:
Tension Coefficient Method for Space Trusses:
Let t_AB = T_AB / L_AB be the tension coefficient of member AB (force per unit length).
1. Component Equations:
Tx = t_AB * (xB - xA), Ty = t_AB * (yB - yA), Tz = t_AB * (zB - zA).
2. Joint Equilibrium:
Sigma t_AB * (xB - xA) + X_A = 0; Sigma t_AB * (yB - yA) + Y_A = 0; Sigma t_AB * (zB - zA) + Z_A = 0.
Forms linear equations without trigonometric functions, ideal for computational matrix solving.

3. Structural Equilibrium & Kinematic Compatibility Check:
- External determinacy verified via global equilibrium equations: Sigma Fx = 0, Sigma Fy = 0, Sigma M = 0.
- Internal determinacy satisfied via Maxwell's criterion: m = 2*j - 3.
- Joint node equilibrium confirmed: Vector resultant of all concurrent member forces vanishes at each pin.

4. Civil Engineering Design & Stability Implications:
Compression members must be checked for Euler column buckling (P_cr = pi^2 * E * I / L_eff^2), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

>> Solution to Question 17:

[Problem Statement]: Explain the effect of temperature changes on statically determinate vs statically indeterminate trusses.

while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

[illegible] an apex load.

at each pin.

4. Civil Engineering Design & Stability Implications:
Compression members must be checked for Euler column buckling (P_cr = pi^2 * E * I / L_eff^2), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

>> Solution to Question [illegible]: [illegible] for limiting them.

Vertical wall reaction balances vertical component: R_v = H * tan(theta).

3. Structural Equilibrium & Kinematic Compatibility Check:
- External determinacy verified via global equilibrium equations: Sigma Fx = 0, Sigma Fy = 0, Sigma M = 0.
- Internal determinacy satisfied via Maxwell's criterion: m = 2*j - 3.
- Joint node equilibrium confirmed: Vector resultant of all concurrent member forces vanishes at each pin.

4. Civil Engineering Design & Stability Implications:
Compression members must be checked for Euler column buckling (P_cr = pi^2 * E * I / L_eff^2), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

[illegible]

2. Mathematical Derivation & Step-by-Step Joint/Section Analysis:
Lack of Fit in Trusses:
1. Determinate Truss: Member fabricated too long by delta_0 causes geometric joint distortion but ZERO residual stress.
2. Indeterminate Truss: Forcing the ill-fitting member into place prestresses the entire framework, inducing self-equilibrating internal residual forces.

3. Structural Equilibrium & Kinematic Compatibility Check:
- External determinacy verified via global equilibrium equations: Sigma Fx = 0, Sigma Fy = 0, Sigma M = 0.
- Internal determinacy satisfied via Maxwell's criterion: m = 2*j - 3.
- Joint node equilibrium confirmed: Vector resultant of all concurrent member forces vanishes at each pin.

4. Civil Engineering Design & Stability Implications:
Compression members must be checked for Euler column buckling (P_cr = pi^2 * E * I / L_eff^2), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

>> Solution to Question 34:
[Problem Statement: Discuss the design of transmission line electrical towers as 3D space trusses under wind and broken-wire conditions.

[Detailed Step-by-Step Answer]:
1. Structural Context & Equilibrium Classification:
This analysis addresses internal member forces and structural determinacy for plane truss systems.
Governing criteria: All joints are frictionless spherical pins; loads act exclusively at nodes; members are ideal two-force axial elements carrying pure tension (+T) or compression (-C).

2. Mathematical Derivation & Step-by-Step Joint/Section Analysis:
Geometric Rigidity of Triangles:
By SSS congruence, three rigid bars connected by pins have exactly ONE unique geometry; angles cannot change without altering member lengths.
A 4-bar quadrilateral has 1 internal degree of freedom (mechanism) and deforms under any infinitesimal shear force.

3. Structural Equilibrium & Kinematic Compatibility Check:
- External determinacy verified via global equilibrium equations: Sigma Fx = 0, Sigma Fy = 0, Sigma M = 0.
- Internal determinacy satisfied via Maxwell's criterion: m = 2*j - 3.
- Joint node equilibrium confirmed: Vector resultant of all concurrent member forces vanishes at each pin.

4. Civil Engineering Design & Stability Implications:
Compression members must be checked for Euler column buckling (P_cr = pi^2 * E * I / L_eff^2), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

>> Solution to Question 26:

[Problem Statement]: Derive the force in the vertical tie of a Queen Post truss of span 12 m with two panels of 4 m.

>> Solution to Question 28: [illegible]

4. Civil Engineering Design & Stability Implications:
Compression members must be checked for Euler column buckling (P_cr = pi^2 * E * I / L_eff^2), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.

>> Solution to Question 30: [illegible] roof and bridge trusses.

5. Member Sizing: Design tension members for net rupture area (A_net * f_y); design compression members for Euler buckling (AISC / IS 800 column curves).
6. Gusset Connection Design: Size welds and bolts at each node.

3. Structural Equilibrium & Kinematic Compatibility Check:
- External determinacy verified via global equilibrium equations: Sigma Fx = 0, Sigma Fy = 0, Sigma M = 0.
- Internal determinacy satisfied via Maxwell's criterion: m = 2*j - 3.
- Joint node equilibrium confirmed: Vector resultant of all concurrent member forces vanishes at each pin.

4. Civil Engineering Design & Stability Implications:
Compression members must be checked for Euler column buckling (P_cr = pi^2 * E * I / L_eff^2), while tension members are governed by net cross-sectional yield strength under bridge/roof design standards.
